Learning reusable control from noisy sparse experiments

Astra research investigation

1 November 2026

Finite acquisition, sharp simultaneous-intervention order,
and certified later decisions

Result and scope

This investigation constructs a finite noisy acquisition method for a whole family of passive responses, derives the asymptotically necessary number of controls that must be activated together, realizes the obstruction with ordinary resistor networks, and implements a retained-record decision certificate. The goal is to make experiments produce reusable knowledge for later interventions, including combinations absent from calibration.

The strongest result is mathematical. It closes a specific finite-noise gap in the pinned Astra determinant representation. The generic experimental budget can be enormous, the numerical demonstration has three controls, and historical priority and a practical advantage over the best equally informed methods are unestablished. The new contribution is a source-specific acquisition and decision guarantee. Rational fitting, linear programming, network energy reduction, and convex optimization are established methods and are credited below.

All prior Astra premises are bound to repository `harjassand/astra-research-knowledge`, revision

`51ce47ce39f976517d60e2f87bbe280b651f6056`.

The complete ordinary proofs, source copies, runnable programs, exact certificates, comparison notes and informative failures accompany this report.

1 The consequential question

An unknown passive system has many possible interventions. Can a finite bank of imperfect scalar experiments support reliable choices after prices, budgets or available actuators change? A complete microscopic model is one possible route. The present route acquires the response law directly, retaining uncertainty in every model consistent with the observations.

The scientific difficulty is operational. A control with zero individual effect can matter in a combination. Small signals can make direct internal-parameter identification unstable. A numerical fit at measured settings does not alone bound an untested combination, and accurate prediction does not automatically supply an efficient optimizer. All three issues are addressed, under a specific fixed-source contract.

Alternative openings were screened before concentrating here. Reversible rare-event lifetime prediction has strong existing potential-theoretic and spectral methods. Broad no-reset quantum claims already have strong subsystem and correlated-noise constructions, including relevant Astra premises. The selected response problem had a concrete missing bridge: N1152 already gave a positive determinant representation and proposed a linear-programming outer family, but left finite noisy convergence unproved. No prior result is reopened or downgraded by this choice.

2 Source and physical operation

There are n finite controls. The same fixed source returns gain

$$F(u) = g^\top P(I + PBP)^{-1}Pg, \quad P = \operatorname{diag}(\sqrt{u}), \quad B \succeq 0, \tag{1}$$

for $u_i \geq 0$, $\sum_i u_i \leq \delta$. Warranted bounds are $|g_i| \leq G$ and $B_{ii} \leq \kappa$. The learner receives neither B nor g. Put $L = \delta G^2$, $q = \delta\kappa$, $x = u/\delta$ and $f(x) = F(\delta x)/L$ on $\mathcal{X} = \{x \geq 0 : \sum x_i \leq 1\}$. Then $0 \leq f \leq 1$. Zero L is trivial. Singular B and exactly dark $g_i = 0$ are permitted.

This contract has a literal passive-network supplier. Clamp two terminals of a resistor network at unit and zero voltage. Complete the unknown baseline Dirichlet energy around its equilibrium voltages. Adding nonnegative conductance u_i across actuator edge i gives terminal conductance

$$J(u) = c_0 + \min_z \left\{ \|z\|^2 + \sum_i u_i(g_i + b_i^\top z)^2 \right\}, \qquad B_{ij} = b_i^\top b_j. \tag{2}$$

At unit voltage, terminal current equals the numerical value of $J(u)$. Eliminating z gives (1). The maximum principle supplies $G = 1$. A known baseline parallel conductance at least $a > 0$ across each controlled edge supplies $B_{ii} \leq 1/a$, hence $q \leq \delta/a$, without reconstructing the hidden network. Those baseline elements have wiring, power and size costs.

For gain tasks, $F(u) = R(u) - R(0)$ for the actual scalar report R; separate unknown offsets need not be identified. The noisy zero reading and its shared error remain in the retained record. The source class, fixedness, available positive doses, calibration, settling and error bounds are premises. A finite fit does not certify them. General chemistry, changing devices, new controls and retained dynamical memory are outside this theorem.

3 Finite noisy acquisition of the entire law

The inherited N1152 determinant identity is

$$f(x) = \frac{N(x)}{D(x)}, \quad D(x) = \det(I + C\operatorname{diag} x), \quad N(x) = \det(I + (C + hh^\top)\operatorname{diag} x) - D(x), \tag{3}$$

where $C = \delta B$ and $h = g/G$. Both polynomials are square-free with nonnegative coefficients. For $s = |S| \geq 1$,

$$d_\varnothing = 1, \quad b_\varnothing = 0, \qquad 0 \leq d_S \leq q^s, \qquad 0 \leq b_S \leq s^2 q^{s-1}. \tag{4}$$

At $q = 0$, the first-degree numerator bound is one and all higher terms vanish.

Choose a retained degree $r \geq 1$ and let $\rho = \min(r, n)$. The inherited one-sided truncation bound is

$$0 \leq f - \frac{N_r}{D_r} \leq e_r, \quad A_r = \sum_{j=0}^{\rho} \frac{q^j}{j!}, \quad e_r = \begin{cases} 0, & \rho = n, \\ q^\rho/(\rho! A_r), & \rho < n. \end{cases} \tag{5}$$

The same truncated coefficients work for the entire simplex.

Independently choose a lattice degree $k \geq 1$. Acquire intervals $f(x_\alpha) \in [y_\alpha - \xi_\alpha, y_\alpha + \xi_\alpha]$, $\xi_\alpha \leq \xi$, at

$$x_\alpha = (\alpha_1, \ldots, \alpha_n)/k, \quad \alpha_i \in \mathbb{N}_0, \quad \sum_{i=0}^{n} \alpha_i = k. \tag{6}$$

There are $M = \binom{n+k}{k}$ settings; each activates at most k controls. The interval errors may be correlated. A single simultaneous coverage event is required.

Form a rational coefficient polytope $\mathcal{P}$ from (4), retaining degrees at most ρ, and impose

$$(y_\alpha - \xi_\alpha - e_\alpha)D(x_\alpha) \leq N(x_\alpha) \leq (y_\alpha + \xi_\alpha)D(x_\alpha). \tag{7}$$

Here $e_\alpha = e_r$ is valid, or use the sharper actual-dose tail in the full proof. All constraints are linear in the coefficients. The true truncated pair is feasible. Every fitted denominator is at least one. A feasible pair need not itself be physically realizable or concave; that relaxation is explicit.

Define the rational finite cross-tail and the simplex norming constant

$$B_{r,k}(q) = \sum_{\substack{1\le a\le \rho,\ 0\le b\le \rho \\ a+b>k}} \frac{aq^{a+b-1}}{(a-1)!b!}, \qquad C_k = \binom{2k-1}{k}. \tag{8}$$

Theorem 1 (Uniform finite-noise acquisition)**.** *On the simultaneous node-coverage event, every pair of feasible responses satisfies*

$$\sup_{x\in\mathcal{X}} \left|\frac{N}{D} - \frac{N'}{D'}\right| \le \Delta_{r,k} := C_k A_r^2(2\xi + e_r) + (C_k+1)B_{r,k}. \tag{9}$$

Consequently

$$f(x) \in \left[\min_{\mathcal{P}} \frac{N(x)}{D(x)},\ \max_{\mathcal{P}} \frac{N(x)}{D(x)} + e_x\right] \cap [0,1] \tag{10}$$

has width at most $\min(1, \Delta_{r,k} + e_r)$*, simultaneously for all future commands, including adaptively chosen commands.*

Proof. For two feasible pairs let $H = ND' - N'D$. At each lattice node, both ratios belong to the same interval of width at most $2\xi + e_r$. The coefficient boxes and the elementary-symmetric bound $\sum_{|S|=j} x_S \le 1/j!$ give $1 \le D, D' \le A_r$. Thus $|H(x_\alpha)| \le A_r^2(2\xi + e_r)$.

The positive degree-a numerator and degree-b denominator terms obey $N_a(x) \le aq^{a-1}/(a-1)!$ and $D_b(x) \le q^b/b!$. Each positive product ND' and $N'D$ therefore has total-degree-$> k$ tail in $[0, B_{r,k}]$. Their signed difference has absolute tail at most $B_{r,k}$, with no extra factor two.

Write $H = H_{\le k} + H_{>k}$. The inherited degree-k simplex interpolation formula has cardinal functions $L_\alpha(x) = \prod_{i=0}^n \binom{k\lambda_i(x)}{\alpha_i}$, with $\lambda_0 = 1 - \sum x_i$ and $\lambda_i = x_i$. Absolute falling factorials are bounded by rising factorials. Their generating functions show $\sum_\alpha |L_\alpha(x)| \le C_k$, independently of n. Hence

$$|H_{\le k}(x)| \le C_k\{A_r^2(2\xi + e_r) + B_{r,k}\}.$$

Add the tail and divide by $DD' \ge 1$. The true truncated pair is feasible, and (5) adds one one-sided remainder to the physical interval. □

Every concrete quantity here is rational for rational specifications. The interval extrema are ordinary linear-fractional programs, transformed to LPs by Charnes–Cooper. Small individual coefficients are retained as uncertainty; they are not recovered by unstable division.

4 A sharp law for simultaneous control order

Taking $r = k < n$ gives

$$B_{k,k}(q) \le \sum_{j\ge k} \frac{(2q)^j}{j!} + q \sum_{j\ge k-1} \frac{(2q)^j}{j!}. \tag{11}$$

For fixed $q > 0$, factorial decay defeats $C_k < 4^k$. A target normalized width $\tau \to 0$ is attained with

$$k = (1 + o(1))\frac{\log(1/\tau)}{\log\log(1/\tau)}, \qquad \xi = \tau^{1+o(1)}. \tag{12}$$

For fixed n, the full-degree alternative $r = n, k = 2n$ eventually suffices without a truncation tail. The order in (12) concerns the largest number of controls activated simultaneously, not the total number of settings.

A matching obstruction built from ordinary resistors

Use $n + 1$ hidden nodes indexed $0, \dots, n$, $n \ge 2$. Each is anchored to the two boundary rails with total conductance $s = 2/\kappa$. Its baseline voltage is a_j, implemented by conductances sa_j to the high rail and $s(1 - a_j)$ to the low rail. Choose

$$a_0 = 0, \quad a_1 = \cdots = a_{n-1} = 1/2, \qquad a_n = 0 \text{ in A}, \quad a_n = 1 \text{ in B}.$$

Actuator i adds conductance between adjacent hidden nodes $i-1, i$. All conductances are nonnegative. Both baseline currents are $c_0 = (n-1)/(2\kappa)$, and the grounded baseline matrix is sI.

If any actuator is off, its missing edge separates the final endpoint from the first. Reflect the voltages of the final component through $v \mapsto 1-v$. The middle-node anchors are symmetric, while the final anchor switches rails. Dissipation is unchanged. Thus *every proper-support terminal-current experiment agrees exactly*, at every amplitude, including adaptive experiments.

For equal full dose $t = \delta/n$, the reduced B has diagonal κ and neighbor entries $-\kappa/2$. The endpoint source entries are $(1/2, \dots, -1/2)$ or $(1/2, \dots, 1/2)$. A tridiagonal cofactor gives

$$\Delta_n = J_B(t\mathbf{1}) - J_A(t\mathbf{1}) = \frac{(\kappa/2)^{n-1} t^n}{D_n} \geq \frac{4L}{n} e^{-q} \left(\frac{q}{2n}\right)^{n-1}, \quad L = \delta/4, \tag{13}$$

because $D_n = \det(I + tB) \leq e^q$.

Theorem 2 (Necessary interaction order)**.** *For fixed $q, L > 0$, any method that uniformly predicts every admitted source to error ε, in every control dimension, with success probability greater than $1/2$, requires maximal simultaneous support*

$$k \geq (1 - o(1)) \frac{\log(L/\varepsilon)}{\log\log(L/\varepsilon)}.$$

This already holds for the resistor networks above.

Proof. Use $n = k + 1$. All permitted transcripts have the same law for A and B. If $\Delta_n > 2\varepsilon$, the two sets of successful predictors are disjoint, so both cannot have success probability greater than $1/2$. The negative logarithm of the lower bound in (13) is $n \log n + O_q(n)$, yielding the result. □

The abstract dark-path construction is inherited from Astra. The quantified accuracy consequence, its matching to the acquired upper bound, and the literal two-rail realization are developments of this investigation. The lower bound does not prove that $\binom{n+k}{k}$ total settings are necessary. A dense background counts as part of an experiment's actual support. Internal voltage probes or extra forcings would change the information interface.

5 Turning the record into an actual decision

Equation (2) makes the true response concave. It does not make every relaxed rational fit concave. The consumer must preserve this distinction. The full proof supplies two routes.

Uniform-value route. A single retained $\widehat{f}$ with $\left\|\widehat{f} - f\right\|_\infty \leq \zeta$ supports any later objective $f(x) - c^\top x$ over any nonempty compact convex $K \subseteq \mathcal{X}$ with a paid linear optimization oracle. The generic source has directional curvature $\mathcal{C} = 8q(1+\sqrt{q})^2$. Finite differences toward the $n+1$ simplex vertices give an approximate Frank–Wolfe direction with loss $\nu = \mathcal{C}h/2 + 4\zeta/h$. After T updates,

$$\text{regret} \leq \nu + \frac{2\mathcal{C}}{T+2}.$$

Choosing $h = \min(1, \varepsilon/(2\mathcal{C}))$, $\zeta \leq \varepsilon h/16$, $T \geq \max(1, \lceil 4\mathcal{C}/\varepsilon \rceil)$ gives regret at most ε. The cost is at most $(n+2)T$ offline evaluations and T linear-oracle calls, with no new source measurements. The $q = 0$ affine case uses one linear optimization. Numerical errors must be paid explicitly.

Sharper task certificates at full retained degree. When $r = n$, a proposed action x has LP-certified value interval $[f_-, f_+]$. Since $D_i(x) \geq 0$,

$$\frac{N_i - f_+ D_i}{D} \leq \partial_i f(x) \leq \frac{N_i - f_- D_i}{D}. \tag{14}$$

Extremizing these linear-fractional quantities over the same coefficient polytope gives $g_- \leq \nabla f(x) \leq g_+$. True concavity and nonnegative coordinates imply

$$\text{regret}(x) \leq \max_{y \in K} (g_+ - c)^\top y - (g_- - c)^\top x. \tag{15}$$

An ordinary nonlinear optimizer may propose x. Its output is accepted by (15), even if its surrogate is nonconcave or its solve is local.

Floating LP duals are proposals only. For transformed constraints $Az \le 0$, $e^\top z = 1$, $0 \le z_j \le U_j$, any rational $\lambda \ge 0, \nu$ proves

$$c^\top z \le \nu + \sum_j U_j \max\{c_j - (A^\top \lambda)_j - \nu e_j, 0\}.$$

Every reported bound is checked with exact fractions. The independent checker needs only the scalar records, public bounds and saved duals.

6 Executed example and informative failure

The main demonstration emulates the four-node network above with the last node attached to the high rail, $\kappa = 4$, $\delta = 1$, $G = 1/2$, $L = 1/4$. Its normalized model has $q = 4$, $h = (1, 0, 1)$ and tridiagonal C with diagonal 4 and neighbor entries -2.

The learner uses 84 degree-six settings, including the measured zero command; 168 nonzero setting writes, 84 resets and corresponding clearing are charged. The assumed calibrated raw-current radius is 1/8,000,000 per setting. The shared baseline subtraction yields normalized radius 10^{-6}. The retained full-degree rational family has 14 coefficient variables.

Later task	Selected normalized allocation	Certified regret, rounded up
All controls	$(0.4166650, 0.1666705, 0.4166643)$	2.20×10^{-5}
Changed prices	$(0.4709220, 0.2565947, 0.2724832)$	2.82×10^{-5}
Control 2 unavailable	$(0.4999999, 0, 0.5000000)$	1.08×10^{-5}
Half budget	$(0.2499999, 0, 0.2500000)$	9.20×10^{-6}

Every task uses zero additional source measurements. Exact fractions are saved. The first two cases miss the earlier 2×10^{-5} normalized target. Certificate validity passes all four; the data were not retuned. Physical objective units multiply these bounds by $L = 1/4$. This is a deterministic calibrated-interval emulator, not a fabricated circuit or a Monte Carlo confidence claim.

The independent validator checks all 84 raw-to-normalized transfers and 32 exact dual certificates without loading B, g, the hidden circuit or true-objective diagnostics. Other exact programs check 4,572 rational norming/tail cases and, for the circuit lower bound, 1,002 proper-support equalities, 2,166 literal Kirchhoff-versus-reduced equalities, and 81 dense gaps. The ordinary proofs carry the general claims; these checks supplement them.

This run uses $r = n = 3$, not a large-dimensional truncation. Its generic uniform width bound is $388157/750000 \approx 0.51754$. The small numbers in the table are *task regrets*, not uniformly tiny error over the whole simplex. The separate algebraic fixture and held-out floating diagnostics are retained with their different scopes.

7 Costs and the strongest conventional alternatives

The settings count is $\binom{n+k}{k}$ and coefficient storage is $2\sum_{j=1}^{\rho}\binom{n}{j}$. Exact rational LP has polynomial cost in this explicit input size and bit length; that statement does not make the input small. Every setting, clearing, settling interval, repeated read, calibration, stored coefficient and future optimization call is part of the bill.

The sufficient generic bounds are already severe at moderate effective dose. For normalized width 0.01, the exact conservative construction with $r = k$ gives the following budgets; these are not physical lower bounds.

q	Sufficient k	Sufficient per-node radius
0.1	5	1.62×10^{-5}
1	25	5.35×10^{-18}
3	68	2.08×10^{-45}

At $n = 1000, q = 1, k = 25$, the setting count is approximately 8.90×10^{49}. The actual LP envelopes can be much narrower, as the task example shows. A scalable large-control acquisition advantage has not been earned.

With a bounded conditionally unbiased sensor of range width W, a sufficient repeat count per raw setting is $2W^2 \log(2M/\alpha)/(L^2\xi^2)$, including baseline subtraction. This requires an appropriate fresh-trial or martingale model. Fixed calibration bias, drift and settling error do not disappear under averaging. A scalar two-point family already gives a worst-case statistical lower bound of order $\varepsilon^{-2}\log(1/\alpha)$; no such lower bound is claimed for a noiseless or calibrated analog interface.

Rational fitting is established. Multivariate interval-rational approximation was developed by Salazar Celis, Cuyt and Verdonk [4]. Cuyt and Yang [5] already derived rational error formulas through polynomial interpolation of a denominator-multiplied discrepancy. Modern positive denominator fits [6] are strong ordinary alternatives. Neither multiplying denominators nor solving a rational LP is a new general algorithm here.

Known-model network optimization is established. Effective-resistance design and related network allocation problems have strong convex methods [7]. For one task, direct noisy convex optimization may require far less calibration than acquiring an entire law. If internal probes or a well-conditioned parametric source model are available, identification can also win. Giving those extra observations to only one comparator would change the problem.

The defensible increment is the explicit uniform finite-noise guarantee for this positive determinant class, including dark and nearly dark controls, its matching simultaneous-support law, and the complete retained-record decision construction. An ordinary rational fitting or design method can adopt these ingredients. No exclusive computational advantage is implied by the name of the learned object.

Scientific conclusion

There is a sharp relationship between required precision and how many controls must be calibrated together, even in simple passive networks. The retained response can be acquired with finite noisy data and used for later certified decisions without identifying weak internal couplings. This is a completed foundational result at the declared interface. The unresolved consequential step is reducing total experimental and precision costs on large, physically qualified systems; the present work does not turn that remaining burden into an assumed service or a civilization-changing claim.

Sources and reproducibility

The archive includes the three complete pinned Astra sources, their exact paths and hashes, ordinary new proofs, scoped internal challenges, all programs and receipts, and a short run guide. Accepted foundations remain accepted at their actual scope. External references are listed rather than republished.

1. Astra N1152, *Positive determinant representation and factorial intervention tails*; pinned corpus, 29 October control-acquisition source. Exact path, complete source and SHA-256 are preserved in the accompanying proof.
2. Astra N1150, *Simplex law acquisition*; the same pinned source tree. Exact path, complete source and SHA-256 are preserved in the accompanying proof.
3. Astra, *Simultaneous intervention order and the precision of a retained response law*; same pinned tree, imported `DARK_INTERVENTION_ORDER.md`.
4. O. Salazar Celis, A. Cuyt, B. Verdonk (2007), *Rational approximation of vertical segments*, Numerical Algorithms 45, 375–388. doi:10.1007/s11075-007-9077-3.
5. A. Cuyt, X. Yang (2010), *A practical error formula for multivariate rational interpolation and approximation*, Numerical Algorithms 55, 233–243. doi:10.1007/s11075-010-9380-2.
6. J. Chok, G. M. Vasil, *Rational Function Approximation with Normalized Positive Denominators*, SIAM Journal on Scientific Computing. doi:10.1137/24M1632139; arXiv:2310.12053, version 4 (2025).
7. A. Ghosh, S. Boyd, A. Saberi (2008), *Minimizing Effective Resistance of a Graph*, SIAM Review 50, 37–66. doi:10.1137/050645452.
8. A. Charnes, W. W. Cooper (1962), *Programming with linear fractional functionals*, Naval Research Logistics Quarterly 9, 181–186. doi:10.1002/nav.3800090303.